White Paper

FPGA

AI on Agilex™ 5 FPGA

This paper presents the deployment of a neural network image classifier on an Altera® Agilex 5 FPGA using the ONE WARE Studio toolchain. It demonstrates a reliable and efficient path from model training to on-device usage.

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Executive Summary

Running a neural network on an FPGA offers key advantages over CPU- or GPU-based solutions: low latency, low power consumption, and direct integration with sensor hardware. This white paper presents El Camino GmbH's deployment of a neural network image classifier on the Altera Agilex 5 FPGA using the ONE WARE Studio toolchain. The design combines an AI engine, a MIPI CSI-2 camera interface, and a video processing path within a single FPGA fabric. The system classifies images across ten object categories with an accuracy of 97.55 % and a latency of 11 ms at a 50 MHz clock. This paper covers the complete workflow, from model training and hardware integration to validation, and provides resource usage and performance data that confirm the suitability of ONE WARE-based AI on Altera®FPGAs.

1 Introduction

Deploying a neural network at the edge places strict demands on hardware: low latency, energy efficiency, and tight integration with sensors. FPGAs are well suited to these requirements. Unlike GPUs or CPUs, they allow the AI to be co-designed with the surrounding I/O elements, without the overhead of a general-purpose operating system. Altera's Agilex 5 is a mid-range FPGA that combines adaptive logic modules, dedicated DSP blocks, and high-speed transceiver interfaces. This makes it well suited for inference workloads that must also handle real-time sensor data, such as live camera streams. El Camino GmbH uses the ONE WARE Studio IDE to generate a trained image classification model for the Agilex 5. ONE WARE covers model training, quantisation, and IP core generation within a single environment, reducing the engineering effort compared to conventional RTL-based flows. The video processing path is configured and the AI results are read back via a Python script using Tcl commands, enabling fast and reproducible operation. This paper documents the system architecture, the development workflow, the FPGA resource footprint, and the measured classification performance.

2 Tooling and Methodology

2.1 ONE WARE Studio

ONE WARE Studio is a software development environment that covers the full AI deployment workflow with its extension ONE AI. In this project, ONE WARE is used to train the image classification model, generate the corresponding AI engine IP core, and export it for integration into the Quartus Prime project. The tool performs model quantisation and a produces a synthesisable IP core without requiring interaction with an external deep learning framework.

2.2 Quartus IDE

The overall FPGA design is implemented in Altera Quartus Prime 25.3. Platform Designer is used to assemble the individual IP cores into a connected block design, manage the interconnect, and configure each IP core's parameters. Quartus Prime handles synthesis, place-and-route, timing analysis, and bitstream generation for the Agilex 5 target device.

2.3 Python and Tcl Automation

The configuration of the video processing path IP cores is handled by a Python script that issues Tcl commands. The script configures the relevant IP cores, and reads back the classification results from the AI engine.

3 Implementation

3.1 System Architecture

The design combines a video preprocessing path for the AI engine and a live video path for display output. Both paths share the same camera input and run concurrently, as shown in Figure 1. The AI path transforms the raw camera image into a format suitable for the neural network. The live video path forwards the unprocessed image to an HDMI display output.

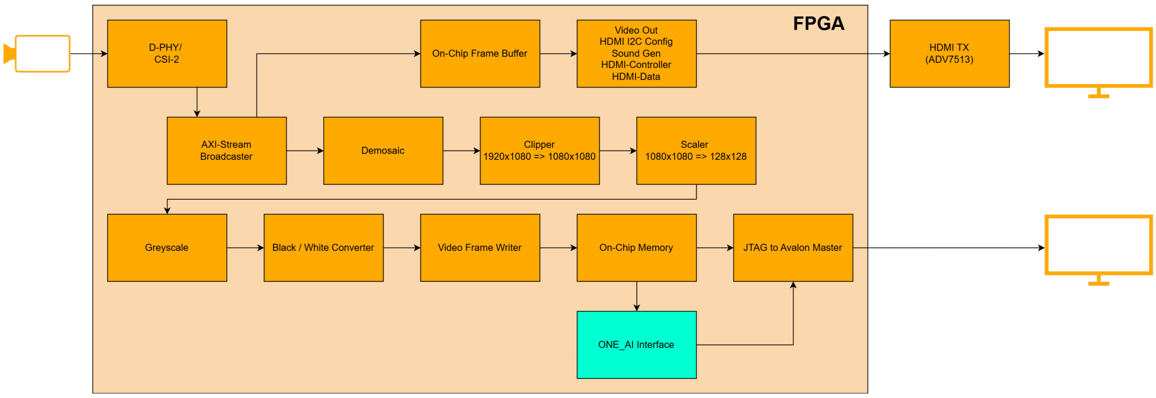

Figure 1: Block design of the AI on Agilex 5 FPGA system

3.2 Camera Input

The camera stream enters the FPGA through a D-PHY / CSI-2 receiver IP core, which processes the MIPI signal and presents a parallel pixel stream. An AXI-Stream Broadcaster then distributes this stream to both the video pre-processing path and the live video path simultaneously.

3.3 Video Pre-processing

The video pre-processing path prepares the camera input for the AI engine. The raw Bayer-pattern output from the broadcaster is passed through a Demosaic IP core, which reconstructs a full-colour image from the sensor's colour filter array. A Clipper then crops the frame from 1920$\times$1080 to a square 1080$\times$1080 region, removing the horizontal margins introduced by the camera aspect ratio. A Scaler downsamples the cropped frame to 128$\times$128 pixels, matching the fixed input resolution of the neural network. The scaled image passes through a Greyscale converter, which reduces it to a single color channel, followed by a Black/White Converter that produces a binary pixel representation. Finally, a Video Frame Writer stores the processed frame in on-chip memory. It also issues an interrupt signal to indicate that a new frame is ready for inference.

3.4 AI Engine

The classifier is trained on a labelled image dataset covering ten object categories: Banana, Cactus, Car, Carrot, Castle, Cat, Cruise Ship, Lightning, Traffic Light, and Umbrella. The trained model is quantised and exported in a format ready for direct hardware synthesis, reducing on-chip memory requirements without loss of accuracy. The ONE_AI Interface reads the preprocessed frame from on-chip memory and passes it to the AI engine, which executes the complete neural network without external processor involvement. All network weights reside in on-chip M20K block RAM, eliminating off-chip memory latency. The engine operates at 50 MHz and the classification result is read out via a JTAG-to-Avalon Master bridge, making the output accessible to the host during operation.

3.5 Live Video Path

The live video path provides a real-time display of the unmodified camera feed. Frames from the AXI-Stream Broadcaster are written directly to an On-Chip Frame Buffer. The buffered output is then processed by a set of video output IP cores, covering HDMI I^2C configuration, HDMI controller, and HDMI data formatting, which drive the external ADV7513 HDMI transmitter to produce the monitor output.

3.6 FPGA Resource Utilisation

Table 2 shows resource consumption broken down to each subsystem. The video pre-processing path needs the majority of logic and registers, driven primarily by the frame buffer and the AXI-Stream infrastructure. The on-chip buffer for the AI engine holds the largest share of block RAM, to store the image data. The AI engine itself is relatively lightweight in terms of logic (Table 1), leveraging the Altera® AI tensor blocks, as the neural network is designed to be compact and efficient.

Table 1: AI Engine Resource Utilisation

Resource	AI Engine
ALMs	3650
M20K Blocks	105
DSP Blocks	65

The total resource usage fits comfortably within the Agilex 5 device, leaving significant capacity for future extensions or a smaller device.

Table 2: FPGA Resource Utilisation by Subsystem

Resource	AI Path	Video Path	MIPI CSI-2 & Config	Total
ALMs	5149	10688	1765 + 530	18132
M20K Blocks	353	77	14	444
DSP Blocks	65	36	0	101

4 Results

4.1 Inference Latency

The timing measurements characterise the latency of the AI engine under operational conditions. Table 3 lists the raw clock cycle counts and the corresponding elapsed times at 50 MHz. The measurement of the clock cycle starts after the interrupt signals that a new frame is ready, and ends when the classification result is available for readout. The setup is designed for a one-shot measurement, predicting one frame and waiting until the result is read out before starting the next inference. The AI engine's output frequency is configurable according to the user's design specification. In this case, the design is configured to produce results for a frequency of 60 fps. The measurements are consistent at 11 ms, representing the deterministic latency for a single classification. The timing of 11 ms is well within the 16.67 ms budget for 60 fps operation, confirming that the design meets real-time inference requirements.

Table 3: Inference Timing Measurements (clock: 50 MHz)

Measurement	**Value**
Amount of Clock cycles	728.262
Time (ms)	10.65

4.2 Classification Accuracy

The model is evaluated on a test set of 20 labelled images per class, 204 samples in total. Table 4 shows the metrics. The model reaches an overall accuracy of **97.55 %**. Macro-averaged precision and recall both exceed 97.5 %, and the alignment between macro and weighted averages confirms that performance is balanced across all ten classes.

Table 4: Overall Classification Performance

Metric	**Macro**	**Weighted**
Accuracy	97.55 %	
Precision	97.72 %	97.76 %
Recall	97.57 %	97.55 %
F1-Score	97.57 %	97.58 %

The confusion matrix in Table 5 shows that most classes are classified correctly. Four classes (Car, Castle, Cruise Ship, and Umbrella) have no misclassifications. One sample each from Banana, Lightning, and Traffic Light is incorrectly predicted as Cruise Ship. Cactus loses one sample to Banana, and Carrot loses one sample to Cat.

Table 5: Confusion Matrix (rows = true class, columns = predicted class)

	Banana	**Cactus**	**Car**	**Carrot**	**Castle**	**Cat**	**Cruiseship**	**Lightning**	**Trafficlight**	**Umbrella**
Banana	**19**	0	0	0	0	0	1	0	0	0
Cactus	1	**19**	0	0	0	0	0	0	0	0
Car	0	0	**20**	0	0	0	0	0	0	0
Carrot	0	0	0	**19**	0	1	0	0	0	0
Castle	0	0	0	0	**20**	0	0	0	0	0
Cat	0	0	0	0	0	**20**	0	0	0	0
Cruise Ship	0	0	0	0	0	0	**20**	0	0	0
Lightning	0	0	0	0	0	0	1	**19**	0	0
Traffic Light	0	0	0	0	0	0	1	0	**19**	0
Umbrella	0	0	0	0	0	0	0	0	0	**20**

5 Conclusion

This paper presents El Camino GmbH's deployment of a neural network image classifier on the Altera Agilex 5 FPGA using the ONE WARE Studio toolchain. The design integrates an AI engine, a MIPI CSI-2 camera interface, and a video processing path within a single FPGA fabric, achieving a classification accuracy of 97.55 % across ten object categories at a latency of 11 ms at 50 MHz. ONE WARE reduces the complexity of FPGA-based AI deployment by handling model quantisation, IP core generation, and hardware mapping within one environment. These results confirm that FPGAs are a practical and competitive platform for edge AI inference. They deliver low latency and direct hardware integration that CPU- or GPU-based solutions cannot match in real-time embedded applications. El Camino GmbH is ready to port this approach to customer-specific requirements that need reliable AI inference on FPGA hardware.